

Thermolysin (8764)

WT 序列:

>tr|S8BE21|S8BE21_DACHA Extracellular metalloproteinase OS=Dactylellina haptotyla (strain CBS 200.50) OX=1284197 GN=H072_8764 PE=3 SV=1

MSGAEPPARSAVFAPALVTPDSRVNYEASNDAILRWYDLSRNQAAAKPFFKPRDVAYDAL
KTYSKQFAWNVDLADLVDHRVNSNGAAHSVRFTQKFKEVPVDASEVVVNLDSDGTVYSI
YNGYHYGIPRNLEPKKITVSQREARVLVDRAVQQTYKEYKVESQTLIVYQYHPFVNRPPK
GKGRSDESSLAILRKATEVQEAEFEFPIAGPQDGQYFLAWDFRVLTONPRNYRLLLD
AMTGKFITVQDLAQYASGTARVFDPNPVVISGNATLAHTTAASVLHGYTQVVGVDRLNPA
DGSGKYHLDGSFVEMDEIEAPTFAEPTSATADFTFETDNRAFLDSMTYFHVDRFQNYIQKS
LSLTNVANYNIKIDPQGFDGQDNSHYIGGAITFGEAGGVDPDAQDAMVILHEYGHAIQDNT
NPGFNNPASGVGEGFGDFLSAIYYDDKHTNPSATRGWMMSWDARGGWGGRRYDLSYNF
DDPQYTSLSNNHDTGQLWCTTGFEYLRKLGGDSSYIGTKHFARDLTIKLHLLANFNVP
SKDATTTQMAQQMEAADGNLGGWYGLADKLHQKVIYDTFRRRHLPGYPNKPVDVYINDG
RDGGYGSTSGNDLFTENLWQESYWDTQDIWVKTPYASAAAQAAGSPNDYVEPPVGSTA
YLYARVKNRGTSGSGPVKVRAFSAQPGIGLSWPDDWFEMDASQVAQPSNITPGTGNGIIV
GPFTWTPTEVGHECVFVIAECNQDKATTQVIAPNARVEHSQVLPFDNNAQRNIHPTAAGG
KFNRGFWVRNPDEITRLISLVYESRLPRGWQFTTNLVNQTQIRLGPLDKRWVELVIDQAA
GAEVTNFDPRFTLTVTGLIEDRVIGGMTYYIAPDSAFAPPRKPPGGHGGHGGHGGHGGN
GGHGGHCGCGSSDNKYCFHISSCKEIEDCSEIEVRIRFDNYKRQRGYLGRQRDYFDV

WT 催化域序列:

>tr|S8BE21|S8BE21_DACHA Extracellular metalloproteinase catalytic domain (thermolysin catalytic domain) three calcium ions and one Zn2+

YASGTARVFDPNPVVISGNATLAHTTAASVLHGYTQVVGVDRLNPADGSGKYHLDGSFVE
MDEIEAPTFAEPTSATADFTFETDNRAFLDSMTYFHVDRFQNYIQKSLSLTNVANYNIKIDP
QGFDGQDNSHYIGGAITFGEAGGVDPDAQDAMVILHEYGHAIQDNTNPGFNNPASGVGEG
FGDFLSAIYYDDKHTNPSATRGWMMSWDARGGWGGRRYDLSYNFDDPQYTSLSNNHDT
GQLWCTTGFEYLRKLGGDSSYIGTKHFARDLTIKLHLLANFNVP
SKDATTTQMAQQMEAADGNLGGWYGLADKLHQKVIYDTFR

选择 5 条序列用于实验，关键指标和理由见下表:

ID	RMSD_sd	Category	HBdens	ΔHBdens%	ΔHB%	ΔhSASA%
_8764_4163	0.0087	≤0.01037	2.7926	+10.21	+7.61	-2.35
_8764_8945	0.0092	≤0.01037	2.7131	+7.07	+6.22	-0.79
_8764_4268	0.0220	0.01037-0.02916	2.7502	+8.53	+9.01	+0.45
_8764_1106	0.0104	0.01037-0.02916	2.7283	+7.67	+10.10	+2.26
_8764_7459	0.0353	0.02916-0.03779	2.7455	+8.34	+9.21	+0.80

最终选择 4 条:

>mpnn_8764_4163

TASGKARVFIPNPVVISGDPTLSHTTAQAVLESYTEVVDVTNLNAPDSSGNYHLDGKFVE
MDEIEAPTFTPESSANADFTFPTNNRAFLDSMTYFHVDSFQNYIINSLSLNDVANYKIKIDS
QGMNGADNSAYIGGAIVFGEGGGVPDAQDAMVILHEYGHAIQDYTNPGFDHPKSGVGE
GFGDFLSAIYYHSKHTNPAATRGHMMSWDARGGDGRRYDLTYNFDDPEYTSLTDNHET
GQLWCTTGFKLFLKLGGDSPYPEVVEFARKLTIRLALLANFNIPAKNATVKQMAEQMVAA
DGNLGGFYGLADKKYADIKETFA

>mpnn_8764_8945

TASGKARVFIPNPVVISGDPTLSHTTAQAVIDSYTTTVDVTNLNAPDSSGNYHLDGKFVEM
DEIEAPTFTPESSANADFTFPTSNRAFLDSMTYFHVDSFQNYIINSLSLTDVANYKIKIDSQG
MNGEDNSAYIGGAIVFGEGGGVPDAQDAMVIMHEYGHAIQDYTNPGFDHPKSGVGEGF
GDFLSAIYYHDKHTNPAAVRGRMMSWDATGGDGRRYDLTYNFKDPAYTSLTDNHETGQ
LWCTTGFKLFLKLGGDSPYPEVVEFARRLTIRLALLANFNIPAKNATVEQMAAQIVAADGN
LGGFYGLADKQYADIKETFA

>mpnn_8764_4268

TASGKARVFIPNPVVISGDPTLSHTTAQAEIEKYTETVTVTNLNAADSSGYYHLDGKFVEM
DEIEAPTFTPESSSNADFTFPTDNRAFLDSMTYFHVDSFQNYIINSLSLTNVANYKIKIDPQG
MNGEDNSAYIGGAIVFGEGGGVPDAQDAMVILHEYGHAIQDYTNPGFDHPKSGVGEGFG
DFLSAIYYHDKYTNPAAVRGRMMSWDARGGDGRRYDLTYNFKDPKYTSLTDNHETGQ
LWCTTGFKLYLALGGDSPYPEVQERARRLTIRLALLANFNVPAKNATVEQMAAQIVAADG
NLGGFYGLADKANAALIKATFA

>mpnn_8764_7459

TASGTARVFIPNPVVISGDPTLSHTTAQAVLEKYTETVTVTNLNAPDSSGNYHLDGKFVEM
DEIEAPTFTPESSANADFTFPTDNRAFLDSMTYFHVDSFQNYIINSLSLTNVANYKIKIDPQG
ENGEDNSAYKGGAIIVFGEGGGVPDAQDAMVILHEYGHAIQDYTNPGFDHPKSGVGEGFG
DFLSAIYYHDKYTNPAATRGRMMSWDATGGDGRRYDLTYNFDDPEYTSLTDNHETGQL
WCTTGFRLYLKLGGDSPYPEIVEFARKLTIRLALLANFNIPAKNATVAQMAAQIAAADGNL
GGFYGLADKQYEQIIKDTFA

Chitinase (1476)

野生型全长序列

>tr|S8ANY0|S8ANY0_DACHA chitinase OS=Dactylellina haptotyla (strain CBS 200.50)
OX=1284197 GN=H072_1476 PE=3 SV=1

MATGFRTVGYVVDWAIYGRNFKPQDLKADQFSHILFAFVKINRDTGEIQLADPWADTDIH
WDESWDVPAGVTNVYGIFLQLFKLKKVHRNLKTMLSIGGWYTYSQDGSFAAGASTPEKRD
KFARSAVQMVKDFGLDGDLDWEYPVDATEAANYVDLLRLCRQYLNEANPAFELSIAAP
CGADKIQKLDIPGMDRYLDFWNL MAYDFAGSWSQAAGHASNIFGSTSNPASTEFSFDTAL
RMYSAVNPHKLVVGMPYGRGFANTDGP GKPYQGTGQGNWETGVWDYKNLPLPGSQE
YQDDQLIASYSYDPAQRLMISYDTPHIAELKAKYIMSRGLGGAMWWETSGDKVGAGSLV
QTVIDTFPPKKRTTAAPAKKKVRVKLAQDLSLSTEEQEVR LAFDYFTDPEELGKDIIQSK
DLKKAFSALGFNLSPGEIKEIKETIDPDDEGFIVYELFLEVAAMKMKDRDGKDELDKAFSL
FTGGDDEGPITLQHLQRVAKALNENVTD DTLRDM LREASSGDRNEVNK

野生型催化域:

>tr|S8ANY0|S8ANY0_DACHA chitinase catalytic domain OS=Dactylellina haptotyla (strain CBS 200.50) OX=1284197 GN=H072_1476 PE=3 SV=1

FRTVGYYVDWAIYGRNFKPQDLKADQFSHILFAFVKINRDTGEIQLADPWADTDIHWDES
WDVPAGVTNVYGIFLQLFKLKKVHRNLKTMLSIGGWTYSQDGSFAAGASTPEKRDKFAR
SAVQMVKDFGLDGIDLDWEYPVDATEAANYVDLLRLCRQYLNEANPAFELSIAAPCGAD
KIQKLDIPGMDRYLDFWNLMAFYDFAGSWSQAAGHASNIFGSTSNPASTEFSDTALRMYS
AVNPHKLVVGMPYGRGFANTDGPQKPYQGTGQGNWETGVWDYKNLPLPGSQEYQDD
QLIASYSYDPAQRLMISYDTPHIAELKAKYIMSRGLGGAMWWETSGDKVGAGSLVQTVI
DTFPPK

ID	RMSD_sd	Category	HBdens	ΔHBdens%	ΔHB%	ΔhSASA%
_1476_672	0.0172	≤0.01856	2.8959	+5.04	+2.58	-2.34
_1476_5220	0.0100	≤0.01856	2.8727	+4.19	+1.06	-3.01
_1476_3917	0.0234	0.01856-0.02672	2.9567	+7.24	+3.70	-3.30
_1476_6648	0.0252	0.01856-0.02672	2.9350	+6.46	+0.81	-5.31
_1476_7011	0.0491	0.02672-0.04912	2.9894	+8.43	+3.53	-4.52
_1476_1420	0.0518	>0.04912	2.9111	+5.46	+3.29	-1.11

选择 4 条序列交付实验

>mpnn_1476_672

MRTVGYYVSWAIYGRKFYPRDLKAELFSHILFAFVDINPETGEIVLRDPYADELVRYDESW
DVPEGVENVYGIFKELQELKKKHRNLKTMLSIGGWTFSAGNFRAGAATPEKRENFAS
AVAMVKKYGLDGIDLDWEYPKDAEEAANYVELLKLKREELDKADPSFELSIAAPCFPENI
KVLIDIEGMDKYLDFWNLMAFYDFAGGWSSYAGHASNIFRSEENPKATPLSFDDALKLYSAV
NPEKLVVGMPYGRGFANTSGLGTSYSGVGTGNYEPGIWDYKDLPLSGSTVHVDEKLIAS
YSYNSTTKYMISYDTTEIAKLKAKYIIEKGLGGAMFWETSGDYEGEKSLIQTVVDTFPPK

>mpnn_1476_5220

MRTVGYYVSWAIYGEKFYPRDLKAELFSHILFAFVDIDPETGEIVLRDKYADVEVRYDES
WDVPEGVKNIYGIFKELRELKKKHRDLKTMLSIGGWTFSAGNFAAGAATPEKRETFEAS
AVAMVKEYGLDGIDLDWEYPKDAEEAANYVALLKLKREKLNEADPSFELSIAAPCYPEQI
AVLDIEGMDKYLDFWNLMAFYDFAGGWSSYAGHASNIFKSSSNPKATPLSFDDALKLYSAV
NPEKLVVGMPYGRGFANTSGLGTSFSGVGGGNFEPGIWNYKDLPLPGATVHVDEKLIAS
YSYNSSKKLMISYDTTEIAALKAKYILDKGLGGAMFWETSGDYEGDKSLIQTVVDTFPPK

>mpnn_1476_3917

MRTVGYYVSWAIYGRKFYPKDLKADLFSHILFAFVDIDPETGEIILRDPYADIEIRYDESWD
VPPGVENIYGIFKELQELKKKHRDLKTMLSIGGWTASASGAFRAGASTPEKRENFANS
AVMVKKYGLDGIDLDWEYPQDAEEAANYVELLKLKREKLKEADPSFELSIAAPCYPEQIKVL

DIEGMDKYLDFWNL MAYDFAGGWS DYAGHASNIFKSEENPKATPLSFDDALKMYS AVNP
EKL VVGMP LYGRGFANT SGLGTPFSGVGGGNYEPGIWDYKNLPLSGATVYTDEKLIASYS
YNSTTKYMISYDTTEIAKLKAKYILEKGLGGAMFWETSGDKEGDES LIKTVVDTFPKK

>mpnn_1476_1420

MRTVGYVSWAIYGRNFYPEDLKADLFSHILFAFVTIDPETGEIKLEDPYADIEIRRDESWD
VPPGVENIYGIFKELNKLKEKHRDLKTMLSIGGWYSSSGAFSGASTPEKRKTF AESAVA
MVKKYGLDGIDL DWEYPKNAEEAANYVELLKL CREKLKEANPDFELSIAAPCYPEYIKV
LDIEGMDKYLDFWNL MAYDFAGGWSSYAGHASNIFKSKDNPKATPLSFDDALKMYS AVN
PEKL VVGMP LYGRGFANT SGLGTSYSGVGGGNYEPGIWDYKNLPLPGATVYEDEKLIASY
SYNSTTKLMISYDTTNI AKLKAKYILEKGLGGAMFWETSGDKEGKDSL IQTVVDTFPKK

Beta-1,3-glucanase (9520)

WT 全长序列

>tr|S8A2J8|S8A2J8_DACHA GH16 domain-containing protein OS=Dactylellina haptotyla (strain
CBS 200.50) OX=1284197 GN=H072_9520 PE=4 SV=1

MDPDRSQFSPFVDPITPVG SVTPYSYSGSEPNSNRGSTSALPSIIGPQQRYFHSRRVPKGTVE
KPWLAKTDPKEKWVTIIPLVGIFIGFAVSGYLVYEGYTSVVKHKYCPVLMEDWSTGFDTN
TWTREVEVGGYGNGQFEYTTNTEENSYVQDGVLYIKPTLQDERLIVQDNTINLIKQGICTS
DVPKNCITATNSTNGTIVNPVKSARINTKKSYSIKYGRVEVEAKLPLGKWLWPAIWMLPTD
SIYGPWPRSGEIDIMESRGNDYRYAQGGNDIVSSALHWGPDNAHDAWWMTNVKHRALH
TTYSAGFNNFGVEWSDKYIFTYVNTRLLQVLYTKFEK KSLWDRGQFPLTSSNGSRYVNPW
YDSKGFNAPFDQSFYLMNVAVGGTNGWFEDGIDDKPWVDASPNAKKDFWNARYMWA
ADWDKNDNHAMQVKS VKMWQQCD

WT 催化域序列

>tr|S8A2J8|S8A2J8_DACHA GH16 domain-containing protein catalytic domain OS=Dactylellina
haptotyla (strain CBS 200.50) OX=1284197 GN=H072_9520 PE=4 SV=1

YCPVLMEDWSTGFDNTWTREVEVGGYGNGQFEYTTNTEENSYVQDGVLYIKPTLQDE
RLIVQDNTINLIKQGICTSDVPKNCITATNSTNGTIVNPVKSARINTKKSYSIKYGRVEVEAK
LPLGKWLWPAIWMLPTDSIYGPWPRSGEIDIMESRGNDYRYAQGGNDIVSSALHWGPDN
AHDAWWMTNVKHRALHTTYSAGFNNFGVEWSDKYIFTYVNTRLLQVLYTKFEK KSLW
DRGQFPLTSSNGSRYVNPWYDSKGFNAPFDQSFYLMNVAVGGTNGWFEDGIDDKPWVD
ASPNAKKDFWNARYMWAADWDKNDNHAMQVKS VKMWQQCD

ID	RMSD_sd	Category	HBdens	Δ HBdens%	Δ HB%	Δ hSASA%	PTM
_9520_6276	0.0223	≤ 0.02333	2.5941	0.93	-4.12	-5.00	0.95
_9520_4063	0.0227	≤ 0.02333	2.5628	-0.29	-3.53	-3.25	0.95
_9520_6175	0.0251	0.02333– 0.02946	2.5888	0.72	-1.16	-1.87	0.95
_9520_2749	0.0338	0.02946–	2.7414	6.66	0.79	-5.50	0.95

		0.06266					
_9520_2266	0.0747	>0.06266	2.7379	6.52	3.34	-2.99	0.95
_9520_6316	0.0723	>0.06266	2.7183	5.76	2.20	-3.37	0.96

最终选择 4 条序列交付湿实验

>mpnn_9520_6276

AKYCPVLDEDWSSGFDTSIWKKEVTVGNGNGQFELTTDTSENSYVKDGVLYIKPTLQD
EALITNDNTINLIAAGTCTSSNPADCIDETNSATGSIVNPVKSARISTRDAYSIKYGRVEVEA
KLPGKGLWPAIWMLPKDKKYGPWPASGEIDIMESRGNDSSYPNGGNSVVYSALHWGP
NSAYDATYLYYKSYKHPNSTYSDGFNNFGVEWNDTYIKTYVNTPENNVLEVNFGETLW
NRGKFPAKDSNGNPLTNPWASSPGNNAPFDQEFYLMNVAVGGTNGWFEDGANNKPWK
DSSPTAKRDFWLARDTWAADWTQNDSAAMKVKSVMWQRCD

>mpnn_9520_6175

DKYCLVLDEDWSNGFDTSIWTQEVTVGNGNGQFELTTNTSENSYVKDGVLYIKPTLQD
ESLITNDTTINLIAEGKCTSTNPADCIDETNKATGTIVNPVKSARINTKNAYSIKYGRVEVEA
KLPGKGLWPAIWMLPKDKKYGPWPRSSEIDIMESRGNDSSYPNGGNNVIYSALHWGP
DSAYDATYLYYKSYTHPNSTYSDGFNNFGVEWNDNYIKTYVNTKDNVVLEVNFETGQTL
WERGKFPATDANGNPNTNPWASSSGYNAPFDQEFYLMNVAVGGTNGWFEDGANNKPW
KDSSPTAKKDFWNARDTWAADWSKNDAAAMKVKSVMWQRCD

>mpnn_9520_2749

NKYCPVLEEDWSNGFDTSIWTQEVTVGNGGNGQFELTTNTSTNSYVKDGVLYIKPTLQDP
ALITNDTTINLIAEGICTSSNPADCIDTTNSATGTIVNPVTSARINTKNAYSIKYGRVEVEAK
LPKGKGLWPAIWMLPKDKKYGPWPRSSEIDIMESRGNDSSHPNGGNDVIYSALHWGPNP
AYDATDKYYKSYKHPNSTYSDGFNNFGVEWDDKYIRTYVNTKDNVVLEVVKFENKTLWE
EGEFPAKDANGNPLKNPWADSKNYPNAPFDQEFYLMNVAVGGTNGWFEDGADNKPWKD
SSPTAKKDFWDARDTWAADWSKNDAAAMKVKSVMWQKCS

>mpnn_9520_2266

AKYCPVLEEDWSNGFDTSVWTQEVTVGNGGNGQFELTTNTSENSYVKDGVLYIKPTLQD
EALITTDNTINLIEQGICTSTNPADCIDETNSATGTIVNPVKSARITTKNAYSIKYGRVEVEA
KLPGKGLWPAIWMLPKDKKYGPWPASGEIDIMESRGNDSSHPNGGNDVIYSALHWGPNP
AAYDATYLYYKSYKHPNSTYSDGFNNFGVEWDDTSIKTYVNTKDNVVLNVNFTGKTLWE
RGKFPAKDSNGNTLTNPWADSKGYNAPFDQEFYLMNVAVGGTNGWFEDGADGKPWTD
SSPTAKRDFWNARDTWAADWEKNGEAAAMKVKSVMWQRCD

Subtilisin-like serine protease (4519)

WT 全长序列:

>EPS41579.1 H072_4519 [Dactylella haptotyla CBS 200.50]

MLSKSLVSIFALAGLALAGPIHKVTNAGAPGTIADRYIVVLKKDLSDAAVDAHTSRIASFH
SNIVRDLTGAKSHGVQNQFKFKSTGFKGYSGGFDAATLQQILRSPEVDFVEQDSIMRINVE
QTDSTWGLDRISHQDYAPYTYEYDEAAAGAGTTVYVIDTGIRITHDEFKGSNGTARATW

GFNAVDKSDSDGNHGHGTHCAGTIAGKTYGVSKKAKVVAVKVLSAGGSGSTSGVVNGMN
WVAENATPNFSVASMSLGGGKSTAINAAVDAIFNAGVTIVVAAGNESQDAKNVSPASAPN
AITVGAISSNKIASFSNWGTLIDVFAPGVDVKSAWGTGDDTTTKTISGTSMACPHVAGLA
AYYISAEGGSGADPAAITDKITGNAVTVGQVTGAIKGSPNKIAYNGVA

WT 催化域

>EPS41579.1 H072_4519 Catalytic domain [Dactylellina haptotyla CBS 200.50]

INVEQTDSTWGLDRISHQDYSAPYTYEYDEAAAGAGTTVYVIDTGIRITHDEFKGSNGTA
RATWGFNAVDKSDSDGNHGHGTHCAGTIAGKTYGVSKKAKVVAVKVLSAGGSGSTSGVV
NGMNWVAENATPNFSVASMSLGGGKSTAINAAVDAIFNAGVTIVVAAGNESQDAKNVSP
ASAPNAITVGAISSNKIASFSNWGTLIDVFAPGVDVKSAWGTGDDTTTKTISGTSMACPH
VAGLAAYYISAEGGSGADPAAITDKITGNAVTVGQVTGAIKGSPNKIAYNGVA

ID	RMSD_sd	Category	HBdens	Δ HBdens%	Δ HB%	PTM
_4519_4907	0.00448 (≤ 0.0077)	2.8304	+13.52	+10.63	-2.55	0.96
_4519_2654	0.00601 (≤ 0.0077)	2.7041	+8.46	+6.33	-1.96	0.96
_4519_17831	0.00937 (0.0077-0.0095)	2.7728	+11.21	+6.93	-3.86	0.96
_4519_8482	0.00947 (0.0077-0.0095)	2.6831	+7.61	+4.58	-2.27	0.96
_4519_1648	0.00992 (0.0095-0.0116)	2.8794	+15.49	+10.95	-3.93	0.96
_4519_30145	0.01075 (0.0095-0.0116)	2.7049	+8.50	+4.27	-1.10	0.96
_4519_11117	0.01330 (> 0.0116)	2.7689	+11.06	+10.87	-0.17	0.96
_4519_12753	0.01780 (> 0.0116)	2.7374	+9.80	+6.38	-3.12	0.96

>MPNN_4519_4907

AIVTDPNATWGLDIISHKDYSAPYTFEYDDAAAGAGTTVYVIDTGIRIDHDEFKGANGESR
ASWGYNVAVDDKDEDGNHGHGTHCAGTIAGRITYGVSKKAKVVAVKVLDENGKGTVAQVV
AGMNWVADNATPNFSVASMSLGGPKDDAIDAAVDAIYNAGVTIVVAAGNNSDASNYSP
ASAPNAITVGAIKNNNTIASFSNYGSLIDVFAPGVDVTSAYGRGDTKSTITASGTSMACPH
VAGLAAYYISSTGGSGADPAAIKAKITGNAVKGKVKGDIKGAPNLIANGIA

>MPNN_4519_17831

AIVTDENATWGLDIISHKNYSAPYTFEYDDAAAGKGTTVYVIDTGIRKDHDEFKGADGSS
RASWGYNVAVDDKDEDGNHGHGTHCAGTIAGRITFGVSKKAKVVAVKVLDENGNGTVEQV
VDGMNWVAANATPKFSVASMSLGGPYSEAINAAVDKIYNAGVTIVVAAGNNSADAANYSP
PASAPNAITVGAIKNNNTIASFSNTGPLIDVFAPGVDVTSAYGTGDTTSTITASGTSMACPH
VAGLAAYYISSTGGSGADPAAIKEWITGNAVKGKVKGDIKGAPNLIANGVA

>MPNN_4519_1648

AVVTDENATWGLDIISHENYSAPYTFEYDDAAAGKGTTVYVIDTGIRIDHDEFKGADGTS
RASWGYNVAVDDDEDVDGNHGHGTHCAGTIAGRITYGVSKKAKVVAVKVLDANGNGTEEQV

VAGMNVVAENATPNFSVASMSLGGPKSAAIDAAVDKIYDAGVTIVVAAGNDSADAANFS
PASAPNAITVGAIKNNNTIASFSNTGPLIDVFAPGVDVTSAYGRGDTTSTITASGTSMACPH
VAGLAAYYISSTGGSGADPAAIKAWITGNAVKGKVKGDIKGAPNLIANGVA

>MPNN_4519_11117

AIVTDKNATWGLDIISHENYSAPYTFEYDDAAAGAGTTVYVIDTGIRIDHDEFKGADGTSR
ASWGYNAVDDVDADGNGHGHGTHCAGTIAGRITYGVSKKAKVVAVKVLDANGNGTEQQVV
DGMKWVAENATPNFSVASMSLGGAKSAAIDAAVDDIYNAGVTIVVAAGNNSDDASNYSP
ASAPNAITVGAIKNNNTIASFSNYGPLIDVFAPGVNVTSA YGTGDTHSTITASGTSMACPHV
AGLAAYYISSTGGSGADPAAIKEKIVGNAVKGKVKGDIKGAPNRIANGIA

Aspartyl aminopeptidase (2059)

WT 全长序列(催化域就是全长序列)

>tr|S8ASA1|S8ASA1_DACHA aspartyl aminopeptidase OS=Dactylellina haptotyla (strain CBS
200.50) OX=1284197 GN=H072_2059 PE=3 SV=1

MASSTSKAFAQGLIEFINASPTPFHAVDTAKKTLLKAGFSEIKERDAWAGKVTSGGKYFLT
RNGSSIIAFAVGKLWKPGNPMSIVGAHTDSCALRLKPVSKRTTEGNTFMQVGVESYGGGL
WHTWFDRLDSMAGRAMVRQKDGNIVSKLVKIDKPILRVPTLAIHLDRQEKFEFNKEGQLF
PILGLVAAELNKGAPTAAGDKMEVENTEEKA EKPFAPLTAITERHHPAVVQLISDATDAQPE
DVLDFEMILYDTQPGTLGGLNEEFIFSARLDNLGMTYCSIEALISSLEASDALDKEPGIRIIA
CFDHEEIGSTTAQGADSNLLPAVVRRLSVLSGDKDYDFASVSSTASNSSAADHLFSPTTTY
YEESLIKSFLISADMAHSVHPNYPGKYESHHRPEMNKGTVIKINANARYATNSPGIVLVQE
CAKEAGVPLQLFVVRNDSSCGSTIGPMLSAQMGMRTLDLGNAQLSMHSIRETGGAYDVE
HAIKLFSTFFQRYSDLEPTIFV

ID	RMSD_sd	Category	HBdens	ΔHBdens%	ΔHB%	ΔhSASA%
_2059_237	0.1058	≤0.20901	3.0308	5.73	−3.27	−8.51
_2059_4489	0.2015	≤0.20901	2.9697	3.60	−0.75	−4.20
_2059_8902	0.2440	0.20901–0.24395	2.9988	4.62	−0.19	−4.60
_2059_3883	0.2165	0.20901–0.24395	2.9370	2.46	−5.23	−7.50
_2059_4444	0.3687	0.24395–0.38089	2.9416	2.62	−1.43	−3.95
_2059_2993	0.3931	>0.38089	2.9415	2.62	+1.28	−1.31
_2059_5765	0.7773	>0.38089	2.9211	1.91	−4.94	−6.72

>MPNN_2059_237

APTPEERAYAEDLLAFINASPTPYHAVATLKEMLAAGFKEVSLDEDWTGIVNPGGKYFLT
KDGRALIAFAIGKKYKPGSPVAIVSSHTDSCRLKCLKENSKALSEGKELLLNVEKYGNNGN
WESWKDLPLSLAGYALVRQPDGSVVKLVRLIDEPIPKIPSVPKSLRDSEERPFDEKELKPV
FGKTEEYLGNLPPPPPPPPPPPPPPPPPPPPFRPKPPLLERHWPELVELIAKEVNAKPEDVL
DFDLFLYPMEPARLIGENGELIEAARLDNKACAYCAVKALIESLADPNALDDYSGIRLVALF

DDEEVGGNSETGAASDLLPTVLKRLSQLVSPDFDVDAFLKAHPGAGVGEKAKAPDPNA
YEQMMKNSFLISFDMAPAVDPDAPEEFHPDWQPQLNQGPVIKLDPERRYPTTEPGVKLVR
KCAELAGVPLQEYRARPDRPTGTTDGPLISERTGMLTIDLGIPMLNMHDAREIGGTKDVLY
AIRLLKTFFEKYEELWPTVEV

>MPNN_2059_8902

APTPEQIAYANDLLDFINASPTPYHAVATLVERLKAAGFKEVSLDEDWTGIVNPGGKYFLT
KDGRALIAFAIGKKYKPGSPVAIVSSHTDSCRLKCLKENSKALSEGKELLNVEKYGNNGN
WESWKDLPLSLAGYALVRQEDGSVEKKLVRITEPILKIPSVPKSLRDSDERPFDLEKELKPV
FGETEKYLGNLPPPPPPPPPPPPPPVPPPPPPFRPKPPLEERHWPELVELIAKEVNAKPEDVL
DFDLFLYPMEPARLIGENGELIEAARLDNKACAYCAVEALIKSLADPKALDNYSGIRLVALF
DDEEVGGNSETGAASDLLKTVLERLSKLVPSDFDVEAFLKANPGAGVGEFAKPYDPNAL
EQMMKNSFLISFDMAPAVDPDAPEEFHPDWQPQLNQGPVIKLDPERRYPTTEPGVKLVKA
CAELAGVPLQEYRARPDRPTGTTDGPLISEKTGMLTIDLGIPMLNMHDAREIGGTKDIYYA
VRLLETFFRNYEKLWPKVER

>MPNN_2059_4444

APTPEQLAYAEDLLEFINRSPTPYHAVAELKKLLEAAGFKEVSLDEDWTGVVNPGGKYFLT
KDGRALIAFAIGKKYKPGSPVAIVSSHTDSCRLKCLKENSRALSPGGKELLNVEKYGNNGN
WESWKDLPLSLAGYALVRQADGSVVKLVKEIKEPILKIPSVPKSLRDSDERPFDLENELKP
VFGKTEEYLENLPPPPPPPPPPPPPPVPPPPPPPRPKPLTERHWPELVELIAKEVNAKPEDVL
DFDLFLYPMEPARLAGENKELIEAARLDNKACAYCAVKALIESLADPKALDNYSGIRLVAL
FDDEEVGGNSDTGAASDLLKTVLERLSKLVPSKDFDVEAFLKAEPGAGVGERAKPYDPD
ALAQMMKNSFLISFDMAPAVDPDAPEEFHPDWRPQLNQGPVIKWDPERRYPTTEPGVKLV
EACAKLAGVPLQRYRARPDRPTGTTDGPLISERTGMLTIDLGIPMLNMHDAREIGGTKDV
YYAVKLLATFFRNYEKLWPTVER

>MPNN_2059_2993

APTPEQLAYAEELLDFINRSPTPYHAVATLVELLKAAGFKEVSLDEDWTGVVNPGGKYFLT
KDGRALIAFAIGKKYKPGSPVAIVSSHTDSCRLKCLKENSKALSEGKELLNVEQYGNNGN
WKS WKDLPLSLAGYALVRQADGSVVKLVKEIKEPILKIPSVPKSLRDSDERPFDLEKELKP
VFGKTEEELGNLPPPPPPPPPPPPPTPNPPPPPPYVPKPPLEERHWPKLVELIAKEVNAKPEDV
LDFDLFLYPMEPARLIGENKELIEAARLDNKACAYCAVKALIESLADPNALDNYSGIRLVAL
FDDEEVGGNSSTGAASDLLKTVLERLSKLVPSDFDVEAFLKANPGAGVGEFAKPYNPNA
LEQMMKKSFLISFDMAPAVDPDAPEEFHPDWRPRLNEGPVIKWDPERRYPTTEPGVKLVK
ACA EKAGVPLQEYRARPDRPTGTTDGPLISEKTGMLTIDLGIPMLNMHDAREIGGTKDIY
YAVKLLSTFFKEYETLWPTVER

Beta-glucosidase (4413)

WT 全长序列（包含催化域）

>EPS41687.1 hypothetical protein H072_4413 [Dactylellina haptotyla CBS 200.50]

MSLPSDFLWGFATAAYQIEGGFDQDGRGPSIWDTFSKTPGKVADGKTGDVACDSYNNA
QDVALSSLGAKIYRFSISWPRVIPLGGRDDPVNEKGVQHYVKFVDMLLDAGIIPMVTLYH
WDLPDELDKRYGGLLNKEEFSADFERYARVMFEALPKVKHWITFNEPWCSAVLGYNVGL

HAPGRTSKREVSPGDSSTECWIVGHSLLVAHGKAVKVYRDEFKPTSGGEIGITLNGDWAE
 PWDPEDPADVEACDRKIEFAISWFADPVYFGHYPESMRKQLGDRLPEFTPEEVELIKGSND
 FYGMNHYTANYIKHRDGEPPADFLGNLETGFYSKSGENIGPETQSVWLRPSPIGFRKLLV
 WLSKRYGRPKIYVTENGTSKAKMADGEKEPPFPEILEDHFRCEYFRGYINAMRDAVEDN
 VNVRIYTAWSLMDNFEWAEGYVTRFGVTYVDYEGGQKRHPKKSARVIGDLFKSLIKDN

ID	RMSD_sd	Category	HBdens	Δ HBdens%	Δ HB%	Δ hSASA%
_4413_288	0.0125	≤ 0.01950	3.0597	-2.41	-3.43	-1.05
_4413_2114	0.0158	≤ 0.01950	2.9925	-4.55	-0.23	+4.53
_4413_4818	0.0305	0.01950-0.04347	3.0640	-2.27	-2.20	+0.07
_4413_2123	0.0565	0.04347-0.05990	2.9652	-5.42	-6.75	-1.41
_4413_2975	0.0610	> 0.05990	3.0630	-2.30	-2.69	-0.40
_4413_5820	0.0753	> 0.05990	2.9756	-5.09	-3.31	+1.87

最终选择以下 4 条序列

>MPNN_4413_288

MKLPSDFLWGFATAAYQTEGGRTKDGRGKSIWDVFSaipgrvadgTTNDVACDSYNNA
 QDVALSSSLGAEIYRFSISWPRVIPNGGADDPVNEKGVQHYVDFVDMLLDAGIIPMVTLY
 HWDLPAALWERYGGLLNKEKFSKDFEAYAAVMFEALPKVKIWETFNEPWHIAVYGYGTG
 TLAPGLTSDKSINPTGDSSTDPWIVAHSLLVAHGKAVKVFREKFQPTNGGEIGMTLNADWA
 EPYDPNNPADVAACERYLDFRHGIFADPIYFGRYPEAARELLGDRLPEFTPEEVELIKGSMD
 FYGMNHYTAYYIKAKEGEPAPDDWLGNVEVSFKNKEGKNIGPETEDVWQHPSPEGFRKL
 LVYLSKRYGNPKIYVTENGTSKAKLAEGEKYPPFPENLNDTFRCEYFKGYIEAMRKAVTE
 DNVNVRIYTWWSLMDNFEWNRGYTTTRFGVTYVDFEGGQKRYPKLSAYVIGELFKSLRA
 KP

>MPNN_4413_4818

KKLPSDFLWGFATAAYQTEGGAGKDGRGESIWDVFSkipgrvadgTTHDVACDSYNNA
 QDVALSSSLGAEIYRFSISWPRVIPNGGADDPVNEKGVQHYVDFVDMLLDAGIIPMVTLY
 HWDLPAALWERYGGLLNKEKFSADFEAYAAVMFKALPKVKIWETFNEPWIDIAVWGYAY
 GWLAPGLTSDKSINPTGDSSTDPWIVAHSLLVAHGKAVKTYRENFKPTNGGEIGITLNANW
 AEPYDPNNPADVAACDRYIDFQHIGIFADPIYFGEYPASIRELLGDRLPEFTPEEKELIKGSM
 DFYGMNHYTAYYIKAKEGEPAPDDVLGNVEVSFKDKNGNIGPETEDVWLHPSPPQGFRD
 LLVYLSKRYGNPKIYVTENGTSKAKLPEGAKRPPFPENLNDDFRCEYFKGYINAMRDAVT
 EDNVNVRIYTAWSLMDNFEWNRGYTTTRFGVTYVDFEGGQKRYPKKSAYVIGELFKSLRE
 KP

>MPNN_4413_2123

MKLPSDFLWGFATAAYQTEGGTTKDGRGKSIWDVFSkipgrvadgTTHDVACDSYNNA
 QDVALSSSLGAEVYRFSISWPRVIPNGGADDPVNEKGVAHYVKFVDMLLDAGIIPMVTLY
 HWDLPAALWERYGGLLNKEKFSSEDAAYAAVMFEALPKVKIWETFNEPWIDIAVWGYGTG

WLAPGLTSDKSINPTGDSSTDPWIVAHSLLVAHGKAVKVYREKFQPTNGGEIGMTLYGDW
AEPYDPNNPADVAACDRYIDFQIGLFADPIYKGEYPASIRELLGDRLPEFTPEEVALIKGSM
DFYGMNHYTAYYIKAKEGEPAPGDWLGNVETLFKNKEGKNIGPETEWVWQHPSPEGFRK
LLVYLSKRYGNPKIYVTENGTSKAKLAPGEKRPFPENLNDTFRCEYFKGYINAMREAVTE
DNVNVKIYTWWSLMDNFEWNLGYTTFRFGVTYVDFEGGQKRYPKKSAYVIGELFKSLKE
KP

>MPNN_4413_2975

KKLPSTDFLWGFATAAYQTEGGWTKDGRGESIWDVWSKIPGRVADGTTGEVACDSYNNA
QDVALSSLGAEVYRFSISWPRVIPNGGADDPVNEKGVQHYVDFVDMLLDAGIIPMTLY
HWDLPKALWDRYGGLLNKEKFSDFAAAYARVMFKALPKVKIWITFNEPWHIAVDGYANG
WLAPGLTSDKSINPTGDSSTDPWIVAHSLLVAHGKAVKIYREEFQPTNGGEIGMTLYADNA
EPYDPNNPADVAACQTYLDFRHGIFADPIYFGDYPESAKKLLGDRLPEFTPEEKALIKGSM
DFYGMNHYTAYYIKAKEGEPAPDDRLGNVEVSFKNKEGKNVGPETEWVWQHPSPEGFR
NLLVWLSKRYGNPKIYVTENGTSKAKLPEGAKRPFPENLNDTFRCEYFKGYINAMRDAV
TEDNVNVKIYTWWSLMDNFEWNLGYTTFRFGVTYVDFEGGQKRYPKKSAYVIGDLFKSL
RAKP

Endothelin converting enzyme (8326)

WT 全长序列

>EPS37966.1 hypothetical protein H072_8326 [Dactylellina haptotyla CBS 200.50]

MPENGHTDSTTEQTPLLGRDAEEASEPSFPLRLKNWFSRHFIIVLLTTLIAGLVAALIVITVR
QDSSPPSSPSPPSPSKGVSVCCTPGCVLAAATLLRSMKRYKSVNPCEDFRTFTCEGFD
AVHEIREDQTNVASLQIMSEEGQLVLKRILESPSPSTESLFWTASNPKDIFSKMQDAYGAC
MNETLLGSLGARPLLDLLSQLEDIYPVKKPKGIDTSLTEAIQFLMGVGVGGPIGLGVGADD
KDPDSNVLQVVPFRSFGPLPSKEYYNNTALVSPLYKDTITSVLEALLTKPSESVLSTFRLKDKP
TVMNKTVVDQVIEFEATMARASDPEDANDVTKYYNPRTLSEAAALPEIAIAADIHQNTSG
YKPPKVIVGSPDYLSKSLKILASQDRKTVQSYLAWKVISAAGGSVEDASVKPLLSFRNRL
QGKAPDVKQERWRTCVRVVDSDLSWILSRSFIEAAFSKEAKDLGDQIIMDIKDEFSRKLQN
SEWMSKKVRKLAIKDVHYIRQKIGYPTKSPDVTDPALLKYYSVDVKVSKNTFFQNRLSAT
RSEVAREWAQAGKPVDKDEWGMTAATVNAYYNPPGNEIVFPAGIMQAPVFYDPSIPKYL
YGAFGAVAGHELSHAFDSSGRNYDQNGNYTDWWDEATVSFAFKKKTDCFVDQYHSYTIP
TKDGPLAINRGLTLGENIADAGGLTAAFAWKRREDEKPDPELLPGFEDFTKEQVFFLAYG
LTWCGKHREEEMIQRITYDPHSPDMFRIKGTANSREWREAYDCPVKEPTCELW

WT 催化域序列

>EPS37966.1 hypothetical protein H072_8326 Catalytic domain [Dactylellina haptotyla CBS 200.50]

TPGCVLAAATLLRSMKRYKSVNPCEDFRTFTCEGFDVAVHEIREDQTNVASLQIMSEEGQL
VLKRILESPSPSTESLFWTASNPKDIFSKMQDAYGACMNETLLGSLGARPLLDLLSQLEDI
YPVKKPKGIDTSLTEAIQFLMGVGVGGPIGLGVGADDKDPDSNVLQVVPFRSFGPLPSKEY
YNNTALVSPLYKDTITSVLEALLTKPSESVLSTFRLKDKPTVMNKTVVDQVIEFEATMARAS
DPEDANDVTKYYNPRTLSEAAALPEIAIAADIHQNTSGYKPPKVIVGSPDYLSKSLKILAS
QDRKTVQSYLAWKVISAAGGSVEDASVKPLLSFRNRLQGKAPDVKQERWRTCVRVVDSD

DLSWILSRSFIEAAFSKEAKDLGDQIIMDIKDEF SRKLQNSEWMSKKVRKLAIDKVHYIRQ
KIGYPTKSPDVTDP SALLKYYS DVK VSKNTFFQNRLSATRSEVAREWAQAGKPV DKDEW
GMTAATVNAYYNPPGNEIVFPAGIMQAPVFYDPSIPKYL SYGAFGAVAGHEL SHAFDSSGR
NYDQNGNYTDWWDEATVSAFKKKTDC FVDQYHSYTIPTKDGPLAINGRLTLGENIADAG
GLTAAFQAWKRRDEEKPD ELLPGFEDFTKEQVFFLAYGLTWCGKHREEEMIQRIYTDPHS
PDMFRIKGT TANSREWREAYDCPVKEPTCELW

ID	RMSD_sd	Category	HBdens	ΔHBdens%	ΔHB%	ΔhSASA%
_8326_7634	0.0058	≤0.00673	3.1132	11.35	4.22	-6.41
_8326_4496	0.0058	≤0.00673	3.1118	11.30	7.08	-3.79
_8326_3348	0.0087	0.00673–0.01026	3.1784	13.69	7.09	-5.80
_8326_2758	0.0081	0.00673–0.01026	3.1033	11.00	4.00	-6.31
_8326_1341	0.0132	0.01026–0.01976	3.1521	12.74	7.25	-4.88
_8326_2598	0.0192	0.01026–0.01976	3.0885	10.47	6.78	-3.34
_8326_2422	0.0199	>0.01976	3.1973	14.36	10.26	-3.59
_8326_3035	0.0414	>0.01976	3.1696	13.37	6.80	-5.79

最终选择以下 4 条序列

>MPNN_8326_7634

SAAVEEAAKTLLDAMSPNRDSVNPCEDFRKFTCEGFLATHEIREDETEVATLSLMQEKIDEI
LKEILESFPPTTKSKYWTASDPDKTIFNKMKDAYNACMNTTELLGKLGAKPLLELLAELEKI
FPVERPAGEDTSLTDAIAYLIKIGVPGPIVIGVDADPKDPTKNVLYVAPPRSFG LPSKEYYYKD
AALVARYKATITSVISALLKHPDKEVLSVFRLPDDPSKLNTSIVDEVIAFEAEMAAAAPDPE
TLNDVRKWYNPMTLEEQAALIPQIAIDKIIAEFTNGYKPDYVIVRDPEY LKKLNEILAKAD
RRTVQAYLQWRIIETWGPYVEDPSIAPLLAFRNELEGRAPDVRPERWRTC VKYVDEILSYI
LSREFIEKAFSKEAKERGLEIIEAIKEEFKKLLENSDWMSKEVREQA IKKVDNIKVKIGYPT
KSPNVKDSKDLLDYKDVK VSKNTFFGNILSAVASKIKMWSQAGKPV DKNEWPMTAA
TVNAYYNPRFNEIVFPAGIMQAPVFYDKSIPRYLSYGAFGAVAGHEL SHAFDSSGRYYDQN
GNYKDWWEETKKA FEETACFVEQYHSYTVETPDG PLPINGKLTLGENIADAGGLKAA
FRAWKKENAKNPSELLPGFEDFTLEQVFFLAYGLTWCGKERE EVIRNRILTDPHSPNEFRIK
GTTENSREWREAYNCPNKEPTCDLW

>MPNN_8326_3348

SAAKEKAAKTLLDAMSPNRASVDPCEDFRAFTCEGFLANHEIREDETRVATLDLMQEKIK
EILKEILESFPPTTKSEYWTASDPDTTVFQKMKDAYNACMNTTELLGKLGAKPLLELLAELE
KIFVEPPPGEDTSLTKAISYLIKIGVSGPIIIGVRADPKDPNKNVLYVAPPRSFG LPSKEYYK
DKELVAEYRATITSVITALLKHPDKEVLSVFRLPDNPDELNTAIVDEVIAFEAEMAAAAPDP
ETRNDVRKWYNPMTLAEQAALIPSI AIDDIIAEFTNGYKPEKVIVEDPAYLKALDKILASTP
RATVQAYLQWRVIEKWGPLVEDPSMEPLLA FRNKLEGRKPDVRKERWRTC VDEV DKNLS
YILSREFIEKAFSKEAKERGEEIIEAIKEEFKKLLENSDWMSKEVRELA IKKVDNIKVKIGYP

TKSPNVTDKDLLEYKKNVSKNTFFGNVLSATKSKIEKMWSQAGKPVDKNEWPMTA
ATVNAYYNPAFNEIVFPAGIMQAPVFYDKSIPKYLSYGAFGAVAGHELSHAFDSSGRYYDQ
NGNYKDWDEETKKAFFEEKTQCFVEQYHSYTIETPKGPLPINGKLTGENIADAGGLKA
AFRAWKKENENKPNDELLPGFEDFTLEQVFFLAYGLTWCGKEREKLVNDILTDPHSPNEFR
IKGTTDNSREWREAYNCPNKEPTCELW

>MPNN_8326_1341

SAAEEAAAATLLSAMSPNRASVDPCEDFRAFTCEGFLATHEIREDETEVATLSLMQEKIDEI
LKKILESFPFTTPSKFWTAKDPDTTIFNKMKDAYNACMNTTELLGKLGDEPLLDKLEELQKI
FPVEKPEGEDTSLTDAIYYLIEIGVPGPIDIGVAADPKDPTKNVLYVAPPRSFGPLPSKEYYDD
EELVAKYRDTITSVITALLKHPRKEVLSVFRLPDNPEEMNTSIVDEVIDFEAKMAKAAPDE
ETRNDIRKYYNPMTLEEAQALIPQIAIDDIKKMTNGYKPEYVIVKDPAYLKALNEILASTD
RRTVQAYLQWRIIEKLGOPYVESPIEPLLAFRNMLEGRKPDVRPERWRCTCVKYVDEYLSYI
LSREFIEAAFSKEAKERGEEIIEDIKEEFKLLAESDWMSKEVRELAIKKVDNIKVKIGYPT
KTPNITDAKNLLEYKDVKVSKNNTFFENILSAIKSKIKKNWAQAGKPVDKNEWPMTAATV
NAYYNPAFNEIVFPAGIMQAPVFYDKSIPSYLSYGAFGAVAGHELSHAFDSSGRYYDQNGN
YADWWDEETKKAFFEEKTACFVEQYHSYSIETKNGPLKINGKLTGENIADAGGLLAAFRA
WKKKNAEKPKLLPGFEDFTLEQVFFLAYGLTWCGKYREEYLKNRILTDPHSPNEFRIKG
TTENSREWREAYNCPVKEPTCELW

>MPNN_8326_2422

SAAEEAAAATLLSAMSPNRASVDPCEDFRAFTCEGFLATHTIREDTEVATLSLMQEKIKEI
LKEILESAPTEKSEFWTAEDPDLTVFQKMKDAYEACMNTTELLGKKGAEPLELLRELQKI
FPVEKPEGEDTSLTDAIAYLISIGVPGPIDIGVDADPKDPNKNVLYVAPPRSFGPLPSKEYYDD
EALVAKYRETITSVITALLKHPDKEVLSVFRLPDNPEEMNTSIVDEVIAFEAEMAKAAPDEE
TRNDITKWYNPMTLAEQAALIPQIAIDDIKFTNGYKPDYVIVKDPEYLKKLNEILAKQD
RATVQAYLQWRIIQKWGPYVEDPSIQPLLAFFENLLNGRPPDAKRERWRCTVDYVDEILSYI
LSREFIEAAFSKEAKEEGKRIENIKEEFKELLKNSTWMSKEVRELAIKKVDNIKVKIGYPT
KTPNVKDAKDLLEYKKNVKVDKNTFFGNILSAIKSKINKNWSQAGKPVDKNEWPMTAAT
VNAYYNPRFNEIVFPAGIMQAPVFYDKSIPRYLSYGAFGAVAGHELSHAFDSSGRHYDQN
GNYTDWWDEATKKAFDEKTACFVEQYHSYTVETKDGPLPINGKLTGENIADAGGLKAA
FRAWKKENAKNPKLLPGFEDFTLEQVFFLAYGLTWCGKEREVIVKNRILTDPHSPNEFRI
KGTTANSREWREAYNCPEKEPTCELW